Integration

Integrate the following w.r.t. x

(1) $\int \left(4x^3 - \frac{1}{x} + \sin x - e^x\right) dx$

(2) $\int \frac{(1-3x)^2}{x^3} dx$

(3) $\int \left(\frac{x^2+5x+6}{x^2+2x}\right) dx$

(4) $\int \left(\frac{2+3\sin x}{\cos^2 x}\right) dx$

(5) $\int \left(\frac{4+3\cos x}{\sin^2 x}\right) dx$

(6) $\int \left(\frac{3\sin x}{2\cos^2 x} + \frac{\cos x}{\sin^2 x}\right) dx$

(7) $\int \frac{\cos 2x}{\cos^2 x \sin^2 x} dx$

(8) $\int \left[\sqrt{1+\sin 2x} + \sqrt{\frac{1+\cos 2x}{1-\cos 2x}}\right] dx$

(9) $\int \sin 5x \cdot \sin 6x \, dx$

(10) $\int e^{\sin x} \cdot \cos x \, dx$

(11) $\int \frac{\sin(\log x)}{x} dx$

(12) $\int 2x e^{x^2} dx$

(13) $\int e^{\tan x} \sec^2 x \, dx$

(14) $\int \frac{x^2}{1+x^6} dx$

(15) $\int \frac{e^x(1+x)}{\sin^2(xe^x)} dx$

Integration by parts

(16) $\int x e^x dx$

(17) $\int x^2 e^x dx$

(18) $\int x e^{3x} dx$

(19) $\int x \log x \, dx$

(20) $\int x^2 \log x \, dx$

(21) $\int x \sin x \, dx$

(22) $\int x \cos 3x \, dx$

(23) $\int x \tan^{-1} x \, dx$

Method of partial fraction.

(24) $\int \frac{x}{(x+1)(x+2)} dx$

(25) $\int \frac{x+3}{(x-1)(x-2)} dx$

(26) $\int \frac{2x+1}{(x+1)(x-3)} dx$

(27) $\int_0^1 \frac{x^3-1}{x-1} dx$

(28) $\int_0^1 \frac{x}{x+1} dx$

(29) $\int_0^1 \frac{2}{1+x^2} dx$

(30) $\int_{-4}^{-3} \frac{x}{7+x}\,dx$

(31) $\int_{1}^{3} \frac{2x}{1+x^2}\,dx$

(32) $\int_{0}^{2} \frac{x^2}{1+x^3}\,dx$

(33) $\int_{-1}^{2} \frac{x\,dx}{x^2+3}$

(34) $\int_{1}^{2} \frac{(\log x)^3}{x}\,dx$

(35) $\int_{0}^{\pi/2} \frac{\sin x}{\sin x + \cos x}\,dx$

(36) $\int_{0}^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,dx$

(37) $\int_{0}^{\pi/2} \frac{\sec x}{\sec x + \operatorname{cosec} x}\,dx$

(38) $\int_{0}^{\pi/2} \frac{dx}{1+\sqrt{\tan x}}\,dx$

(39) $\int_{0}^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x} + \sqrt{\tan x}}\,dx$

(40) $\int_{0}^{\pi/2} \log_e(\tan x)\,dx$

(41) Find the area of the circle $x^2+y^2=a^2$.

(42) Show that the area of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is πab

(43) Find the area of the region bounded by the curve $y=x^2$, x-axis and the line $x=2$ and $x=3$

(44) Find the area of the region bounded by $y=2x^2$, X-axis and line $x=5$

(45) Find the area bounded by the curve $y=x^2-7x+10$ and x-axis

(46) Find the area included by the parabola $x^2=4y$ and the line $y=x$

(47) Find the area of the region bounded by the curve $y=x^2$ and $y=x+2$

(48) Find the area included by the parabola $y^2=4ax$ and $x^2=4ay$